ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
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17.8IO Structural Checklist for Building Types S5: Steel Frames with Infill Masonry Shear Walls and Stiff Diaphragms and S5a: Steel Frames with Infill Masonry Shear Walls and Flexible Diaphragms

Table 17-17. Immediate Occupancy Structural Checklist for Building Types S5 and S5a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the reinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3, is less than 70 lb/in.^{2} (0.48 MPa).	5.5.3.1.1	A.3.2.4.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the unreinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3, is less than 30 lb/in.^{2} (0.21 MPa) for clay units and 70 lb/in.^{2} (0.48 MPa) for concrete units. Bays with openings greater than 25% of the wall area shall not be included in A_w of Equation (4-8).	5.5.3.1.1	A.3.2.5.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INFILL WALL CONNECTIONS: Masonry is in full contact with frame.	5.5.3.5.1 5.5.3.5.3	A.3.2.6.1	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad. The anchor rods are capable of resisting the overturning force using the Quick Check procedure of Section 4.4.3.6.	5.7.3.1	A.5.3.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 12 ft (3.6 m) and aspect ratios less than 1-to-1 in the direction being considered.	5.6.2	A.4.2.1	

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C	NC	N/A	U			
☐	☐	☐	☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 30 ft (9.2 m) and aspect ratios less than or equal to 3-to-1.	5.6.2	A.4.2.2
C	**NC**	**N/A**	**U**	BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 90 ft (27.4 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.3
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 15 ft (4.6 m) if unblocked or 25 ft (7.6 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2.5 if unblocked and 1:1.5 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion.	5.6.2	A.4.2.4
☐	☐	☐	☐			

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Low Seismicity)							
Seismic-Force-Resisting System							
C	**NC**	**N/A**	**U**	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
☐	☐	☐	☐				
C	**NC**	**N/A**	**U**	REINFORCING AT WALL OPENINGS: All wall openings that interrupt rebar have trim reinforcing on all sides or are checked as unreinforced infill frames.	5.5.3.1.5	A.3.2.4.3	
☐	☐	☐	☐				
C	**NC**	**N/A**	**U**	PROPORTIONS: The height-to-thickness ratio of the unreinforced infill walls at each story is less than 13.	5.5.3.1.2	A.3.2.6.2	
☐	☐	☐	☐				
C	**NC**	**N/A**	**U**	CAVITY WALLS: The infill walls are not of cavity construction.	5.5.3.5.2	A.3.2.6.3	
☐	☐	☐	☐				
C	**NC**	**N/A**	**U**	INFILL WALL ECCENTRICITY: The centerline of the infill masonry wall is not offset from the centerline of the steel framing by more than 25% of the wall thickness.	5.5.3.5.3	A.3.2.6.5	
☐	☐	☐	☐				

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Connections						
C	**NC**	**N/A**	**U**	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of loads to the shear walls, and the connections are able to develop the lesser of the shear strength of the walls or diaphragms.	5.7.2	A.5.2.1
Diaphragms (Stiff or Flexible)						
C	**NC**	**N/A**	**U**	OPENINGS AT FRAMES: Diaphragm openings immediately adjacent to the braced frames extend less than 15% of the frame length.	5.6.1.3	A.4.1.5
C	**NC**	**N/A**	**U**	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7
C	**NC**	**N/A**	**U**	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
Flexible Diaphragms						
C	**NC**	**N/A**	**U**	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2
C	**NC**	**N/A**	**U**	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced concrete consist of horizontal spans of less than 40 ft (12.2 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1
C	**NC**	**N/A**	**U**	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1
Connections						
C	**NC**	**N/A**	**U**	STIFFNESS OF WALL ANCHORS: Anchors of concrete or masonry walls to wood structural elements are installed taut and are stiff enough to limit the relative movement between the wall and the diaphragm to no greater than 1/8 in. (3.1 mm) before engagement of the anchors.	5.7.1.2	A.5.1.4

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Foundation System						
C	**NC**	**N/A**	**U**	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1

Status				**Evaluation Statement**	**Tier 2 Reference**	**Commentary Reference**	**Comments**
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C	**NC**	**N/A**	**U**	PROPORTIONS: The height-to-thickness ratio of the unreinforced infill walls at each story is less than 8.	5.5.3.1.2	A.3.2.6.2	
Connections							
C	**NC**	**N/A**	**U**	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation, and the anchorage is able to develop the least of the following: the tensile capacity of the column, the tensile capacity of the lowest-level column splice (if any), or the uplift capacity of the foundation.	5.7.3.1	A.5.3.1	

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